

CSE-214 Online

Behavioural Design Pattern

Duration: 30 Minutes

Problem: Hospital Emergency Room Coordinator

A hospital emergency department handles patients through several medical units. For this problem, consider the Doctor, Pathology Lab, and Radiology Unit. A patient is initially examined by the Doctor, who may request a pathology test, a radiology investigation, or both.

Coordination rule: The medical units must not communicate directly with each other. All requests, results, and notifications pass through a central **Emergency Center**, which coordinates the workflow and keeps track of pending investigations.

Patient Investigation Workflow

When a doctor examines a patient, one or more investigations may be requested. For example:

```
Patient ID: P101

Requested Investigations:
- Pathology Test
- Radiology Test
```

The Emergency Center receives the requests and forwards each request to the appropriate unit.

1. Pathology Test

1. The request is forwarded to the Pathology Lab.
2. The Pathology Lab performs the test.
3. When the result is available, the lab submits it to the Emergency Center.
4. A pathology result is reported as NORMAL or CRITICAL.

2. Radiology Investigation

1. The request is forwarded to the Radiology Unit.
2. The Radiology Unit performs the investigation.
3. When the result is available, the unit submits it to the Emergency Center.
4. A radiology result is reported as OK or NOT OK. For this problem, NOT OK is treated as an urgent result that requires immediate notification.

Result Coordination

The Emergency Center must maintain, for each patient, which investigations were requested and which results have already arrived.

Case 1: Only One Investigation Requested

If only one investigation was requested, the Emergency Center forwards the result to both the Doctor and the Patient as soon as it becomes available. The investigation is then considered complete.

Case 2: Both Investigations Requested

If both pathology and radiology were requested, the Emergency Center waits until both results are available before sending the complete investigation summary to the Doctor and the Patient.

```
Pathology Result Received
Radiology Result Pending

-> Investigation is still incomplete.

Pathology Result Received
Radiology Result Received

-> Complete investigation is ready.
```

Urgent Result Handling

An urgent result must be reported immediately. This occurs when the Pathology Lab reports CRITICAL or the Radiology Unit reports NOT OK.

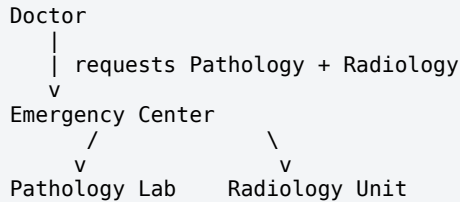
The Emergency Center must immediately notify both the Doctor and the Patient, even if another requested investigation is still pending. However, the overall investigation remains incomplete until every originally requested investigation has finished.

Pathology Result: CRITICAL
Radiology Result: Pending

URGENT: Critical pathology result found for Patient P101.
Doctor notified immediately.
Patient P101 notified immediately.

Example Scenario

Consider Patient P101. The Doctor requests both a pathology test and a radiology investigation.



Suppose the pathology result arrives first and is CRITICAL while the radiology result is still pending. The Emergency Center immediately alerts the Doctor and the Patient, but keeps the investigation open.

Pathology Result: CRITICAL
Radiology Result: Pending

Later, the Radiology Unit reports:

Radiology Result: OK

Now all requested investigations are complete, so the Emergency Center sends the complete result set to both the Doctor and the Patient.

Possible Output

Pathology test requested for Patient P101.
Radiology investigation requested for Patient P101.

Critical pathology result received for Patient P101.
URGENT notification sent to Doctor.
URGENT notification sent to Patient P101.

Radiology result received for Patient P101.

All requested investigations completed for Patient P101.
Complete results sent to Doctor.
Complete results sent to Patient P101.

Implementation Requirements

- The Doctor, Pathology Lab, and Radiology Unit should not hold direct references to one another for workflow communication.
- All investigation requests, result submissions, urgent alerts, and completion notifications should be coordinated through the Emergency Center.
- The Emergency Center must track requested, pending, and completed investigations separately for each patient.
- The design should make it possible to add another investigation unit later with minimal changes to the existing medical units.

You may introduce any necessary classes, interfaces, and methods to produce a clean object-oriented solution.